

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Examination (CE)

Dec 2012

CE702: Advanced Computer Network**Date: 19.12.2012, Wednesday****Time: 10:00 am To 01:00 pm****Maximum Marks: 70****Instructions:**

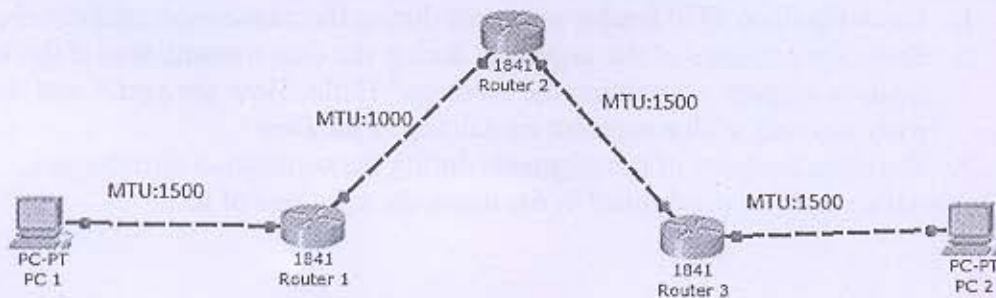
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I**Q - 1 Write the answer of following Questions. [11]**

- (a) Convert the following IP multicast addresses to Ethernet multicast addresses. How many of them specify the same Ethernet address? [2]
1. 224.18.72.8
 2. 235.18.72.8
 3. 237.18.6.88
 4. 224.88.12.8
- (b) TCP receives a segment with destination port address 234. TCP checks and cannot find an open port for this destination computer. So ICMP has to reply with error message. Write down the value of fields for the ICMP message generated. [2]
- (c) How does host can use their IPv4 addresses embedded into IPv6 addresses? [2]
- (d) How does multicast IP addresses delivered over data link layer? [2]
- (e) Why is there a restriction on the generation of an ICMP message in response to a failed ICMP error message? [3]

Q - 2 Write the answer of following Questions. [12]

- (a) By considering following network topology, explain fragmentation and reassembly procedure for sending of 5MB of data from PC1 to PC2. [5]



- (b) Write and explain, types of error reporting messages of ICMPv4. [7]
- OR**
- (b) By taking suitable example, write and explain working of OSPF protocol. [7]

Q - 3 Write the answer of following Questions. (Attempt any TWO) [12]

- (a) What are the responsibilities of ARP and RARP protocol? Draw and explain ARP frame format.
- (b) Write and explain options supported by IPv4 header format.
- (c) An ISP is granted a block of addresses starting with 120.60.4.0/20. The ISP wants to distribute these blocks to 100 organizations with each organization receiving 8 addresses only. Design the sub-blocks and give the slash notation for each sub-block. Find out how many addresses are still available after these allocations.

SECTION – II

- Q - 4 Write the answer of following Questions.** [11]
- (a) What are the types of errors handled by ICMPv6? [2]
 - (b) In Charusat Campus, one wants to download two files named as file.doc (2KB) & assignments.pdf (5MB) from Central server ftp://172.16.0.2. What is role played by control connection and data connection during accessing ftp server? [2]
 - (c) What is the size of a query message requesting the IP address for challenger mail.charusat.ac.in? [2]
 - (d) A DHCP packet is encapsulated in a UDP packet, which is encapsulated in an IP packet, which is in encapsulated in an Ethernet frame. Find the efficiency of a DHCP packet when no option is used. The efficiency in this case should be measured into the number of bytes in the DHCP packet to the total number of bytes transmitted at the physical layer. [2]
 - (e) What are the reasons for development of IPv6 protocol, although IPv4 was very successful? [3]
- Q - 5 Write the answer of following Questions.** [12]
- (a) How does name-address resolution carried out in DNS system? [5]
 - (b) Draw and explain TCP State Transition diagram. [7]
- OR**
- (b) Draw and explain TCP sender and receiver side Finite State Machine (FSM). [7]
- Q - 6 Write the answer of following Questions. (Attempt any TWO)** [12]
- (a) What are the types of TFTP messages? TFTP uses UDP protocol to transfer file, How does it manage flow and error control?
 - (b) TCP opens a connection using an initial sequence number 14,000. The other party opens the connection with an ISN 21,111.
 - 1. Show the three TCP header segments during the connection establishment
 - 2. Show the contents of the segments during the data transmission if the initiator sends a segment containing the message "Hello, How are you?" and the other party answers with a segment containing "Fine Dear".
 - 3. Show the contents of the segments during the connection termination.
 - (c) What is silly windows syndrome? Write down the solutions of it.